Improving Printing Stability of 1pL Inkjet Head for Mass Production of High-Resolution Display Panels

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Abstract
This paper reports on the technology for improving printing stability. We focused on 1pL printhead design to suppress nozzle drying and misting. We confirmed that landing accuracy of 1pL head didn't deteriorate over 16days without any recovery work and the 1pL head could successfully print on a 400ppi pixel bank.

Author Keywords
1pL inkjet head; printing stability; nozzle drying; misting; ink diffusion; single-drop ejection velocity; resonance period

1. Introduction
Display panels require higher resolution when the panel size is medium and small [1][2][3]. At SID2025, Panasonic reported on the technology for printing G8.5 350ppi OLED panels using inkjet printing [4]. This year's report describes technologies for improving the printing stability of 1pL inkjet head. Inkjet heads with small droplet size such as 1pL have a risk in printing stability since the small nozzles can promote the nozzle drying, and the lightweight of droplet can promote misting. It indicates that printing stability is the key technology to success in mass production of high-resolution display panels.

To improve printing stability, we focused on the printhead design to apply critical issues shown in Figure 1. The main issue is to prevent nozzle drying by expanding the nozzle size through the printhead design to shorten the resonance period. The others are to promote ink diffusion and to prevent misting. Although these issues can be approached from multiple angles, including printing process and ink formulation, we believe the foundational performance is governed by the printhead design. Therefore, we focused on printhead design, especially on the nozzle, flow structure and PZT actuator.

2. Design to Expand the Nozzle

2.1. Nozzle Expansion via Resonance Shortening
The volume ejected from the nozzles of the inkjet head is estimated by the relationship in equation (1).

$$V = (S \times v) \times T/2 \times \alpha \qquad (1)$$

V: droplet volume *S:* nozzle cross section v : ejection velocity
T : resonance period α : coefficient

To achieve droplet miniaturization, four approaches can be considered. [A] reducing the ejection velocity can deteriorate landing accuracy due to the influence of airflow. [B] decreasing the nozzle size can increase the risk of nozzle drying and particle adhesion. [C] adjusting the ink properties to shorten the ligament is effective, but relying too much on ink properties alone may lead to performance degradation. Consequently, we thought [D] shortening the resonance period is considered the most efficient approach, to achieve small droplet ejection while expanding the nozzle size.

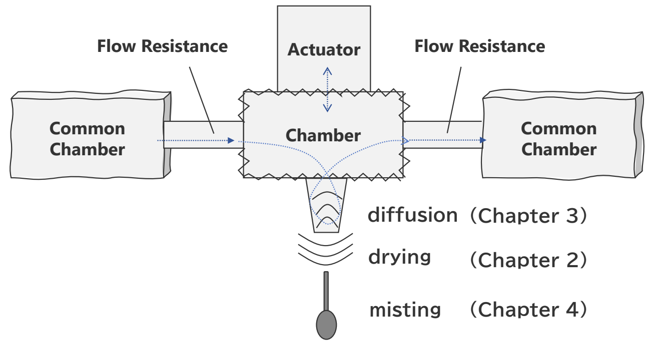

Figure 1. Internal Structural Model of the Inkjet Head

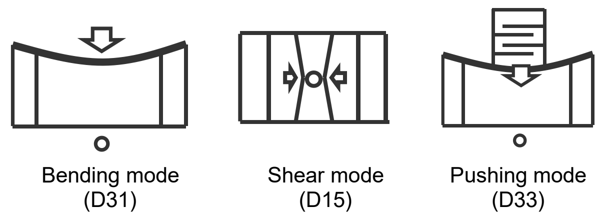

Figure 2. Representative Actuation Methods

To shorten the resonance period of the inkjet head, it is well known that reducing the compliance of the chamber shown in Figure 1 is the most effective method [5]. To lower the compliance of the chamber, Reducing the size and increasing the rigidity of the chamber is also known to be effective, and many solutions are invented [6]. However, reducing the size of the chamber decreases the pressing area of the actuator, requiring a larger displacement from the actuator. Increasing the rigidity of the chamber walls requires stronger force from actuator. To accomplish them at the same time, printhead design in a comprehensive manner including the flow structure and the actuator mechanism is needed. Figure 2 shows a representative actuating mechanism of an inkjet head. The bending mode (D31) can generate relatively large displacement, but there is difficulty to generate strong force. The shear mode (D15) can produce stronger force, but it has limitations to simultaneous ejection. We adopted pushing mode (D33) as a method that can generate both large displacement and strong force with extended durability. Although the displacement is not as large as the bending mode, it can be amplified by stacked PZT while maintaining stiffness and high density, with advanced fabrication technologies. By utilizing these features, we aimed for a design of small and rigid chambers to shorten the resonance period.

Figure 3 shows the experimental result about relative changes in nozzle diameter against the resonance period as a parameter, under the condition of 1pL droplets ejection. It was confirmed that the resonance period and nozzle diameter exhibit an inverse proportional relationship, almost consistent with theoretical predictions. Even with the same type of ink, the nozzle diameter could be expanded by 30%, and by applying it to a different type of ink, the nozzle diameter could be expanded by 40%.

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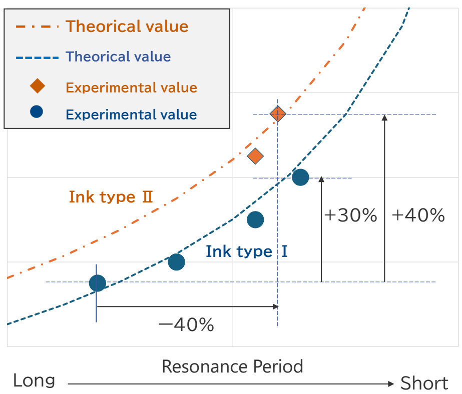

Figure 3. Correlation between nozzle and resonance

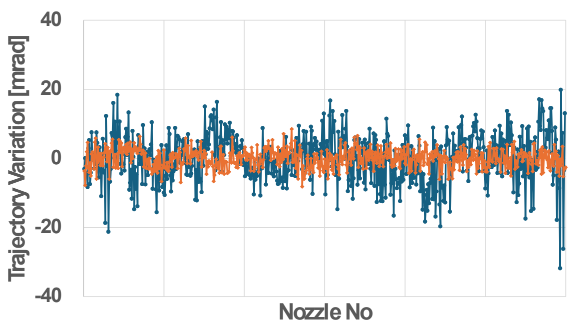

Figure 4. Trajectory variation

Figure 4 shows an experimental result that compares the trajectory variation before and after nozzle diameter expansion. It can be observed that expanding the nozzle diameter significantly reduces the trajectory variation. This is a ripple effect of nozzle diameter expansion, attributed to the reduced sensitivity to inevitable physical variations in nozzle fabrication.

2.2. Suppression of crosstalk

Short resonance period is achieved by reducing the compliance of the chamber and lowering the flow resistance. However, these approaches can promote pressure transmission to neighboring nozzles, inducing "fluidic crosstalk". Large displacement against narrow pushing area with the D33 pushing mode is achieved by the more stacked PZT layers. It can weaken the stiffness of the printhead, which can enhance structural interactions during droplet ejection called "structural crosstalk". It has a significant impact since droplets as small as 1pL are easily affected by even minimal deformation. Therefore, to achieve the inkjet head design with enlarged nozzle through the approach of shortening resonance period, we needed to address the trade-off: the shorter the resonance period, the more severe the crosstalk becomes.

Fluidic crosstalk was suppressed by design of segmented two-stage orifice structure (Figure 5). The short resonance utilized for ejection is formed by the inner, softer flow resistance, while pressure transmission to other nozzles is suppressed by the outer, stiffer flow resistance. Structural crosstalk was suppressed by preventing deformation, warping and distortion of each element during ejection. These are achieved by enhancing stiffness by treating the base plate, PZT, and flow plate as a single rigid body and by optimizing each dimension and bonding methods for minimal deformation through analysis (Figure 6).

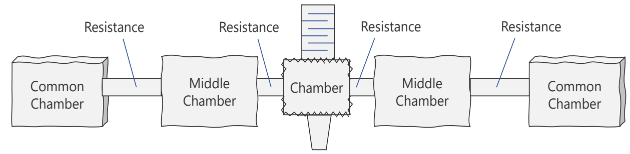

Figure 5. Flow Model to Suppress Fluidal Crosstalk

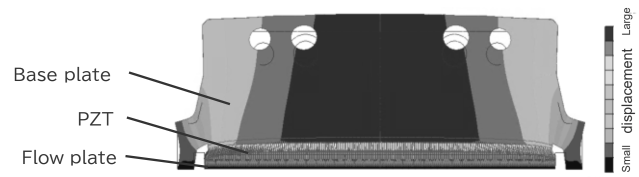

Figure 6. Structural Model to Suppress Structural Crosstalk

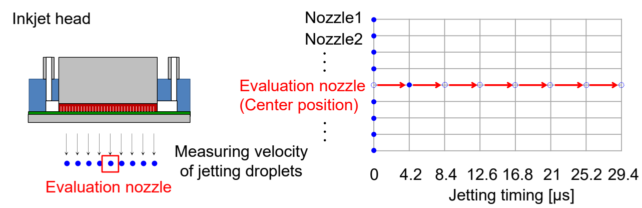

Figure 7. Experimental condition

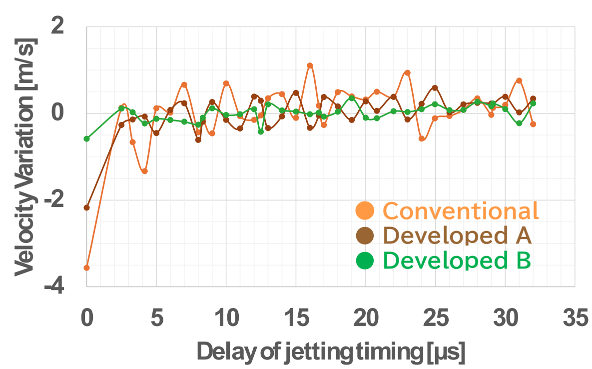

Figure 8. Experimental result

Figure 7 shows the evaluation method for fluidic and structural crosstalk. Droplets were ejected from all nozzles simultaneously, while the ejection timing of the target nozzle was incrementally delayed. The corresponding changes in ejection velocity were monitored to assess the influence of crosstalk. The velocity variation at 0μs indicates structural crosstalk, while the variation observed afterward reflects fluidic crosstalk. Figure 8 shows the evaluation result. Compared to Conventional, Developed A is a printhead with fluidic crosstalk countermeasures implemented. Although a significant change in ejection velocity was observed at a delay time of 0μs, subsequent variations were suppressed. It indicates that although structural crosstalk remains unchanged, fluidic crosstalk has been improved. And the Developed B, which was implemented both structural and further fluidic crosstalk countermeasures, exhibits few variations at a delay time of 0μs and thereafter.

By successfully suppressing crosstalk, which is a critical side effect of short resonance design, it was demonstrated that practical-level ejection accuracy can be achieved while shortening the resonance period and expanding the nozzle diameter.

3. Design to Enhance the Ink Diffusion

Drying from the nozzle tip leads to a localized increase in ink concentration, causing film formation or partial solidification and adhesion to the nozzle. These phenomena are governed by not only "drying rate" but also "diffusion rate". To promote the diffusion rate, printing processes like ink circulation and ink oscillation are effective [7]. In addition to that, we focused on the inkjet printhead design to enhance these printing process effects.

Figure 9 presents the analysis of ink circulation velocity near the nozzle under the same circulation flow rate. It indicates the ink circulation velocity near the nozzle varies depending on the flow structure. It means optimizing the flow structure can enhance the effectiveness of ink circulation process. In this study, although (C), (D) showed significantly improved circulation velocity near the nozzle, they also showed crucial side effects. Therefore, configuration (B), which achieved three times the flow velocity near the nozzle compared to conventional (A), was adopted.

Oscillation effect can be also enhanced through resonance period design. By slightly delaying the resonance period on the upstream relative to that on the downstream, the components diffused by oscillation can be effectively driven downstream. To achieve this, to set the upstream middle chamber and common chamber to be larger or softer, or to increase the flow resistance on the upstream side, shown as Figure 10 are effective.

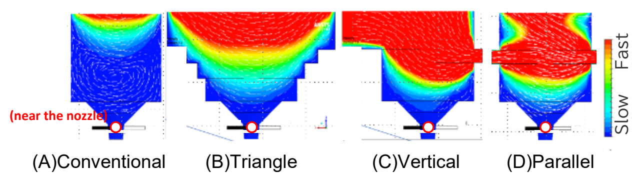

Figure 9. Flow structure to maximize ink circulation process

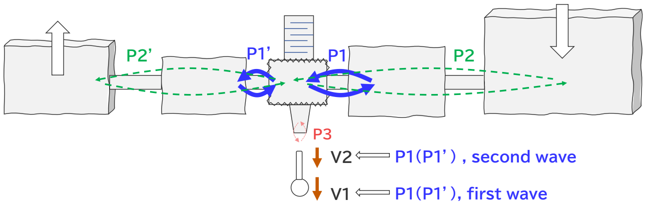

Figure 10. Hypothesis Model

4. Design to Suppress Misting

A droplet ejected from the nozzle may become mist if it is overcome by air resistance or airflow energy. Mist can adhere to the nozzle surface, leading to deterioration in ejection stability. To prevent them, ejecting ink faster as a single drop is effective [8]. Although the velocity of single-drop formation is mainly determined by ink properties and the ejection waveform, we found inkjet head design itself can also contribute. Figure 10 shows the hypothetical model for improving single-drop ejection velocity by giving sufficient energy to the ligament from the energy of the second wave of resonance. To transfer the resonance energy to the ligament, the design to enhance the spectral peak of the resonance by preventing dispersion of the wave is thought to be important. Figures 11 and 12 present the experimental results, aimed at validating the hypothesis.

Figure 11 shows the relative ejection velocity on the vertical axis, and the parameter for waveform (T2/T1) on the horizontal axis. Plots A through E represent different head structures, with the compliance of the middle chamber progressively increased from E to A. When the resonance is pure and its peak is strong, it is theoretically expected that velocity peak will appear at T2/T1 = 1, 3, 5, ..., and that these peaks gradually diminish. This result indicates that the head type is approaching the theoretical value as it moves from type E to type A. It means increasing the compliance of the middle chamber can enhance the purity of the resonance, creating a function of a buffer to the middle chamber and allowing resonance to be confined within the inner region.

Figure 12 shows the relative single-drop ejection velocity to the target value on the vertical axis, and the parameter for waveform(T4/T1) on the horizontal axis. Plots A through E represent the same structural head as Figure 11. Across all plots, single-drop ejection velocity is lowest near T4/T1 ≈ 2, where the damping function becomes theoretically maximized. Although the ejection becomes unstable and useless when the damping function is insufficient, this result supports the validity of the hypothesis that the second resonant wave contributes to enhancing single-drop ejection velocity. It was also observed that printheads with higher resonant purity tend to exhibit an overall improvement in single-drop ejection velocity like plot A, even within the range of stable along the horizontal axis. As a result, by employing the design that enhances resonance purity, we were able to improve single-drop ejection velocity by approximately 10%.

Figure 13 shows the droplet ejection behavior of a 1pL printhead designed to enhance single-drop ejection velocity by increasing resonance purity. The ligament catches up with the leading droplet quickly before landing, resulting in mist-free ejection.

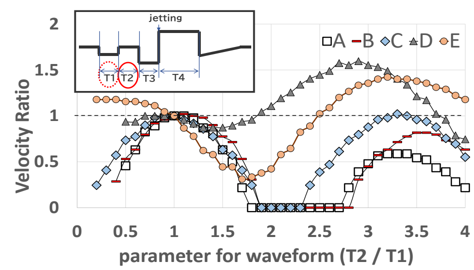

Figure 11. Ejection velocity and T2 (/T1)

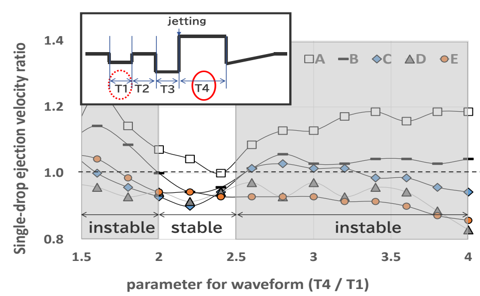

Figure 12. Single-Drop ejection velocity and T4 (/T1)

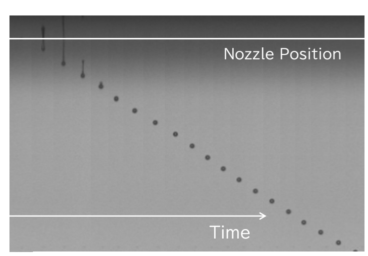

Figure 13. Image of single-drop ejection

5. Printing Evaluation

5.1. Printing Stability Verification

An evaluation of ejection stability of the 1pL inkjet head over time was conducted using a drop watcher system. Ink circulation was maintained with a flow rate of 3 mL/min and a nozzle internal pressure of -0.5 kPa. A process that constantly oscillates the meniscus near the nozzle is applied to stabilize the ejection. This oscillation is called "Tickle". The tickle frequency was set to 1 kHz. The drop watcher system measured the droplet ejection velocity and the variation in ejection trajectory.

If solid remains such as dried ink adhere to the nozzle lip, the effective nozzle diameter becomes smaller, resulting in an increase in ejection velocity and greater variation in ejection trajectory. In this study, no increase in ejection velocity or variation in ejection trajectory was observed over a four-week period, and no clogged nozzles were detected as shown in Figure 14. During the four-week period, no maintenance such as pressure purging or nozzle wiping was performed, only ink circulation and tickle operation were used. The ability to operate without maintenance is a critical factor when applying inkjet technology to mass production, as it contributes to reducing panel manufacturing costs by lowering ink consumption and increasing equipment uptime. In other words, the highly stable results under maintenance-free conditions indicate that the technology is at a practical level for implementation in mass production processes.

We also conducted an evaluation of droplet landing position accuracy using our printing system. Figure.15 shows the time-dependent variation in the repeatability of droplet landing positions along the printing scan direction. Over the 16-day period, droplet landing repeatability remained within 2.0μm, confirming stable printing performance.

5.2. Printing Evaluation on a 400ppi Pixel Bank

Using the 1pL inkjet head previously evaluated for printing stability, we conducted a printing evaluation on custom-fabricated 400ppi pixel bank patterns shown in Figure.16. The ejected volume was adjusted to 0.94pL ±1.7%. The 400ppi resolution pattern had pixel dimensions of 13.0μm × 53.0μm-56.0μm, a bank width of 8.1μm, a bank height of 1μm, and a pixel pitch of 63.3μm. This printing evaluation utilized ink containing dissolved polymer materials and a hydrophobic bank both made by Mitsubishi Chemical Corporation. Printing was performed by repeating the ejection of one drop per scan twice.

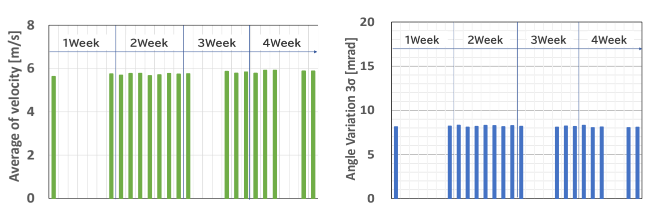

Figure 14. Evaluation Results of Ejection Stability

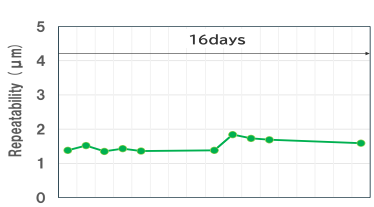

Figure 15. Evaluation Results of Printing Repeatability

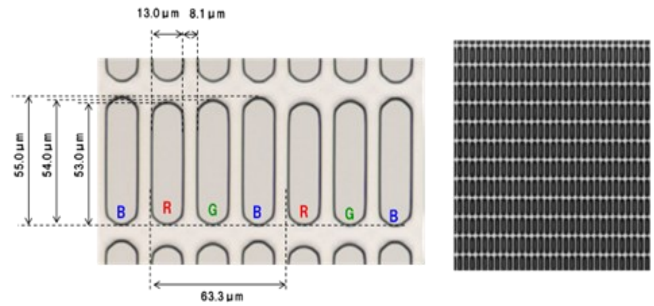

Figure 16. Printing Image of 400ppi Pixel Pattern

Figure.16 shows the printed result. As seen in the image, printing was successfully performed without any color mixing into neighboring pixels. In other words, the 1pL printhead was confirmed not only to have printing stability, but also to achieve printing accuracy for high-resolution panel at the 400ppi level.

6. Conclusion

We developed technologies to enhance the printing stability of 1pL inkjet head, focusing on printhead design for suppressing nozzle drying and misting. Specifically, the printhead design to expand nozzle by 40%, to accelerate the flow velocity near the nozzle by 3times, and to improve single-drop ejection velocity by 10% were developed. In evaluations using the developed 1pL inkjet head, no variation of ejection velocity and trajectory over 4weeks and no deterioration in landing accuracy over 16days were confirmed without any recovery operations. We also successfully confirmed that the 1pL inkjet head can print onto a 400ppi pixel bank without any color mixing.

As a result, we could confirm the possible feasibility of applying inkjet printing systems with 1pL head to mass production of high-resolution display panels. Furthermore, building on this advancement and insight in printhead design, we will aim to develop inkjet technologies for ejecting even smaller droplets for higher resolution displays.

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